Sri Lanka Institute of Information Technology



IT3101 - Data Warehousing and Business Intelligence

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Design and Implementation of a Data Warehouse and Business Intelligence Solution

Group ID: 2026-DS-66

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Domain: Retail / E-commerce
Dataset: E-Commerce Sales Analytics

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1. Dataset Selection and Business Scenario Identification

1.1 Business Domain

Domain: Retail / E-commerce

The selected dataset represents an e-commerce business that manages customer orders and transactions through online sales channels. It contains information related to customers, products, orders, payments, shipping, delivery, returns, marketing, sales, costs, and profitability.

The dataset supports analysis across multiple business dimensions, including time, customer, product, location, sales channel, payment method, marketing campaign, and order status.

1.2 Purpose of the System

The operational system records and manages the day-to-day activities of the e-commerce business. It maintains information related to:

- Customers and customer segments
- Orders and order status
- Products and product categories
- Payments and payment status
- Shipping and delivery
- Returns and cancellations
- Marketing campaigns and coupon usage
- Sales, discounts, taxes, and shipping costs
- Product costs and profitability
- Customer ratings and reviews
- Customer loyalty activities

The primary purpose is to support the processing and management of online sales transactions while maintaining information required to monitor customers, products, sales, and business performance.

1.3 Business Problem

The operational system stores individual transactions and related business activities, making it difficult to obtain a consolidated view of business performance across different areas.

The business therefore requires a centralized Data Warehouse and Business Intelligence solution to analyze sales and profit trends over time, product and category performance, customer segment performance, regional performance, returns and cancellations, marketing campaigns and sales channels, the effect of discounts on profitability, and delivery performance in relation to customer ratings.

The proposed Data Warehouse will integrate the available operational data and support historical, multidimensional analysis for informed business decision-making.

1.4 Dataset Sources

The project uses multiple CSV data sources representing different aspects of the e-commerce operation.

Data Source	Main Purpose
ecommerce_sales_customer_analytics_150k.csv	Main transactional and order-level information
customer_master.csv	Customer master information
product_catalog.json	Product master information
order_items.csv	Product-level details associated with orders
dataset_statistics.csv	Dataset-level statistical information

The primary transactional dataset contains approximately **138,116 records**, while the supporting dataset provides additional customer, product, order-item, and statistical information.

1.5 Dataset Format

All selected data sources are provided in Comma-Separated Values (CSV) format, except for product_catalog.json, which is provided in JSON (JavaScript Object Notation) format. The files can be extracted, validated, transformed, and integrated before being loaded into the analytical environment.

1.6 Dataset Attributes

Group	Attributes	Description
Customer	customer_id, customer_name, customer_age, gender, customer_segment, customer_type, customer_city, customer_state, customer_country, region, customer_postal_code, customer_lifetime_value, is_repeat_customer, customer_order_count	Identifies customers and provides demographic, geographic, segment, and customer behaviour information.
Order & Transaction	order_id, order_date, order_time, order_status, sales_channel, quantity, gross_sales, discount_amount, tax_amount, shipping_cost, net_sales, product_cost, profit, profit_margin_percentage	Records order transactions, their status, sales values, costs, discounts, and profitability.
Product	product_id, product_name, product_category, product_subcategory, brand, supplier, unit_price, product_cost, product_rating	Describes products and their categories, suppliers, pricing, costs, and ratings.
Payment & Delivery	payment_method, payment_status, shipping_method, warehouse, delivery_days, estimated_delivery_days, delivery_status	Records payment details and information related to order fulfilment and delivery performance.
Marketing & Loyalty	marketing_channel, campaign_name, coupon_code, loyalty_points_earned, loyalty_points_redeemed	Captures marketing campaigns, promotional

		activities, and customer loyalty activity.
Feedback & Returns	customer_rating, review_sentiment, customer_review, return_status, return_reason	Records customer feedback, satisfaction, and information about returned orders or products.

1.7 Suitability for Data Warehousing and Business Intelligence

Realistic Operational Environment
The dataset represents a realistic e-commerce environment containing customer orders and related information such as products, sales, payments, delivery, returns, and customer feedback.

Sufficient Data Volume
The primary transactional dataset contains 138,116 records**,** providing sufficient transaction-level data for analytical processing. Supporting customer, product, and order-item datasets provide additional information for analysis.

Multiple Analytical Dimensions
The dataset supports analysis across multiple dimensions, including Time, Customer, Product, Location, Sales Channel, Payment, Marketing, Delivery, and Returns**.** This allows business performance to be analyzed from different perspectives.

Support for Dimensional Modelling
The dataset contains measurable values such as quantity, sales, costs, discounts, and profit, together with descriptive attributes such as customer, product, date, and location information. These can be organized into fact and dimension tables for multidimensional analysis.

Support for Historical Analysis
The presence of order_date allows business performance to be analyzed over time, including revenue, profit, sales trends, returns, and changes in customer behavior.

Support for Business Intelligence
The combination of transactional measures and descriptive attributes supports BI dashboards containing KPIs such as Total Revenue, Total Profit, Total Orders, Average Order Value, and Return Rate, along with interactive analysis by time, region, customer, product, and marketing dimensions.

2. Data Source Identification and Preparation

2.1 Data Sources

The proposed Business Intelligence solution uses multiple CSV data sources representing different aspects of an e-commerce operation. The sources contain transactional, customer, product, and order-item information. These sources can be integrated through their common identifiers to create a unified analytical environment.

Data Source	Format	Records	Purpose
ecommerce_sales_customer_analytics_150k.csv	CSV	138,116	Contains the main order-level transactional and customer-related information.
customer_master.csv	CSV	25,000	Contains master information about customers, including demographic and geographic attributes.
product_catalog.json	JSON	1,175	Contains product information, including categories, brands, suppliers, prices, and ratings.
order_items.csv	CSV	397,569	Contains product-level details associated with individual orders.
dataset_statistics.csv	CSV	1	Contains provides summary-level information about the overall dataset.

2.2 Available Attributes

The main sales source contains attributes related to orders, customers, payments, delivery, returns, marketing, loyalty activity, sales, costs, and profitability. Important attributes include order_id, order_date, customer_id, product-related transaction information, payment_method, payment_status, delivery_status, quantity, gross_sales, discount_amount, net_sales, product_cost, and profit.

The customer_master.csv source contains customer attributes such as customer_id, customer_name, customer_age, gender, customer_segment, customer_city, customer_state, customer_country, region, customer_postal_code, and customer_acquisition_cost.

The product_catalog.json source contains product attributes including product_id, product_name, product_category, product_subcategory, brand, supplier, unit_price, product_cost, and product_rating.

The order_items.csv source contains order and product-level transaction attributes including order_id, product_id, quantity, unit_price, discount_percentage, discount_amount, gross_sales, tax_amount, shipping_cost, net_sales, product_cost, and profit.

The dataset_statistics.csv source provides summary-level information such as total transactions, total customers, total products used, total revenue, total profit, average order value, average rating, return rate, and cancellation rate.

2.3 Relationships Between Data Sources

The sources are connected through shared identifier fields. The main relationships are:

- customer_master.customer_id → ecommerce_sales_customer_analytics_150k.customer_id
- ecommerce_sales_customer_analytics_150k.order_id → order_items.order_id
- product_catalog.product_id → order_items.product_id

The main sales dataset contains one record per order, as it contains 138,116 records and 138,116 unique order_id values. The order_items dataset contains 397,569 records and therefore represents individual product items within orders. Consequently, the sales source operates at order grain, while the order items source operates at order-item grain.

The dataset_statistics.csv file is a summary-level source and does not function as a transactional source. It can be used as a reference for validating overall dataset statistics.

2.4 Source Validation

Before integration, the relationships and key fields were validated to identify duplicate identifiers and invalid references.

Validation Check	Result
Duplicate customer IDs	0
Duplicate product IDs	0
Duplicate order IDs in sales	0
Customer IDs in sales missing from customer master	0
Order IDs in order items missing from sales	0
Product IDs in order items missing from product catalog	0

The validation results indicate that the main identifiers are unique within their respective master datasets and that all customer, order, and product references can be matched across the relevant sources. Therefore, no referential-key correction is required before integration.

2.5 Data Preparation

Data quality checks were performed before the sources were considered for integration. No missing values were identified in the customer master, product catalog, or order items datasets. Missing values were identified in several fields of the main sales dataset.

Attribute	Missing Records	Preparation Approach
delivery_days	24,557	Investigate and handle as unavailable delivery information during ETL.
estimated_delivery_days	24,557	Investigate and handle as unavailable delivery information during ETL.
return_status	128,654	Preserve missing values where the order was not returned.
return_reason	128,654	Preserve as unavailable where no return occurred.

customer_rating	24,557	Treat missing ratings as unavailable rather than assigning an assumed rating.
review_sentiment	24,557	Treat missing review sentiment as unavailable.
customer_review	24,557	Treat missing reviews as unavailable.
campaign_name	83,233	Preserve as unavailable where no campaign information exists.
coupon_code	110,502	Preserve as unavailable where no coupon was used.

The missing values in return-related fields are considered meaningful because return information is only applicable to orders that were returned. Similarly, missing campaign and coupon information may indicate that an order was not associated with a campaign or coupon. Therefore, these values should not be replaced with arbitrary values that could misrepresent the original data.

The order_date field is currently stored as an object/text data type. It will be converted into a proper date format during the ETL process to support time-based analysis. The available order data ranges from 1 January 2021 to 31 December 2025.

Categorical fields such as order status, sales channel, payment method, payment status, delivery status, gender, and customer segment were also examined. The values showed consistent category naming, with no obvious variations caused by inconsistent capitalization or spelling.

2.6 Integration into a Unified Analytical Environment

The validated data sources will be integrated through an ETL process. The CSV files will first be extracted into a staging area, where data types, missing values, formats, and data quality will be checked and prepared.

The prepared sources will then be integrated using their shared identifiers. Customer information will be connected using customer_id, order information using order_id, and product information using product_id. The transformed data will subsequently be loaded into the data warehouse according to the dimensional structure defined for the BI solution.

This integration allows information from different operational sources to be combined into a single analytical environment. As a result, the organisation can analyse sales performance together with customer, product, order, delivery, payment, and profitability information without relying on separate source files.

3. Data Warehouse Architecture Design

3.1 Architecture Overview

The proposed architecture follows a layered Data Warehouse and Business Intelligence structure consisting of four main layers: the Data Source Layer, Data Integration Layer, Storage Layer, and Presentation Layer. The architecture is designed to collect data from the identified e-commerce sources, prepare and integrate the data through an ETL process, store the integrated data in a dimensional Data Warehouse, and provide analytical information through OLAP and Power BI dashboards.

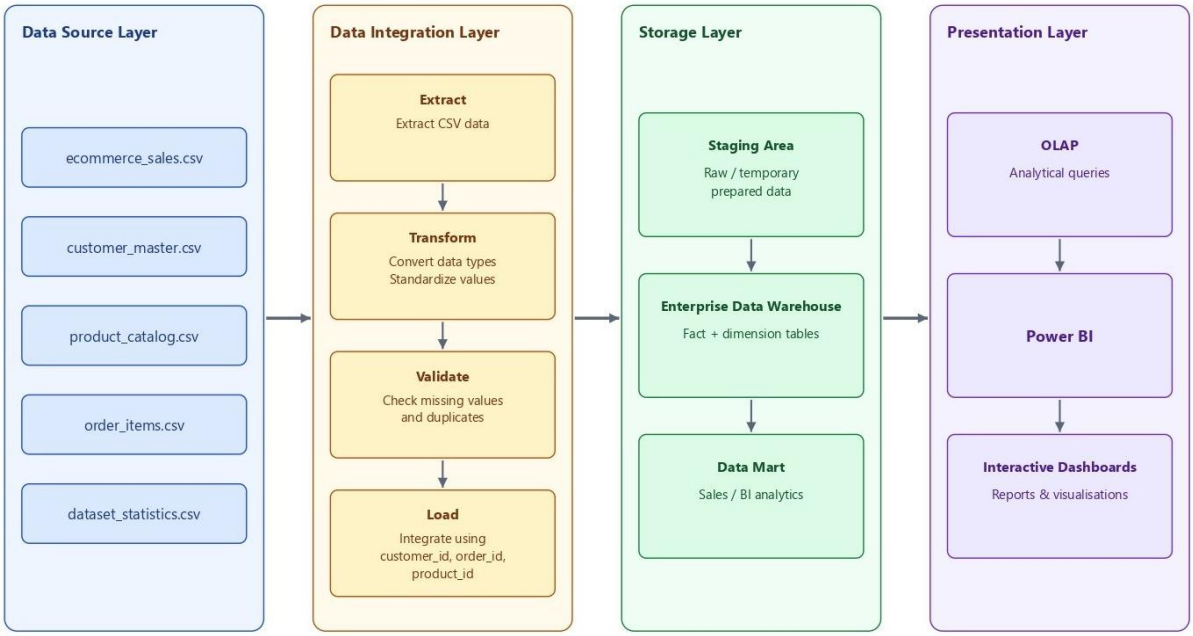


Figure 3.1: Data Warehouse Architecture

3.2 Data Source Layer

The Data Source Layer contains the operational and external data sources used by the BI solution. The selected sources are:

Data Source	Purpose
ecommerce_sales_customer_analytics_150k.csv	Order-level transactional data
customer_master.csv	Customer and demographic information
product_catalog.json	Product and product-category information
order_items.csv	Product-level order details
dataset_statistics.csv	Summary statistics for reference and validation

The main sources are connected using common identifiers such as customer_id, order_id, and product_id.

3.3 Data Integration Layer

The Data Integration Layer performs the Extract, Transform, and Load (ETL) process required to prepare the source data for analytical use.

Extraction: Data is extracted from the selected CSV sources and transferred to the staging environment.

Transformation: The extracted data is cleaned and prepared by handling missing values, checking duplicates, converting data types, standardizing values, formatting order_date, and preparing the data for the dimensional structure.

Relationships are also validated using customer_id, order_id, and product_id. The validation identified no duplicate customer, product, or order identifiers and no missing customer, order, or product references.

Loading: The transformed and validated data is loaded into the Data Warehouse as fact and dimension tables for analytical processing.

3.4 Storage Layer

The Storage Layer consists of the Staging Area, Enterprise Data Warehouse, and Data Mart.

Staging Area: Temporarily stores extracted source data and provides an intermediate area for data quality checks and transformations.

Enterprise Data Warehouse: Acts as the central repository for integrated and historical business data. It uses a dimensional structure consisting of fact and dimension tables to support analysis across areas such as time, customers, products, and locations.

Data Mart: A Sales Analytics Data Mart provides a focused environment for sales and business performance analysis. It contains the data required for business users and BI reporting.

3.5 Presentation Layer

The Presentation Layer provides access to the processed data for analysis and decision-making.

OLAP: OLAP enables multidimensional analysis using operations such as filtering, slicing, drill-down, and roll-up.

Power BI: Power BI is used as the primary BI and reporting tool to connect to the analytical data and create interactive visualizations.

BI Dashboards: The dashboards present KPIs, trends, comparisons, and interactive analysis of sales, revenue, profit, customer, product, and other business performance measures.

4. Dimensional Data Warehouse Design and Implementation

4.1 Business Process

The warehouse supports two related business processes: e-commerce sales analysis, and order fulfilment and customer service analysis. Together they cover sales, revenue, costs, discounts, profitability, customer behaviour, payment methods, sales and marketing channels, geographical performance, delivery performance, customer ratings, returns, cancellations and loyalty programme activity.

- **Retail Sales:** the sale of a product within a customer order, recorded in FactSales at product-line level.
- **Order Fulfilment and Customer Service:** the fulfilment and servicing of a customer order, recorded in FactOrderService at order level.

Keeping the two processes in separate fact tables allows detailed sales analysis without double-counting order-level measures when an order contains several products.

4.2 Fact Tables

The warehouse contains two fact tables that share a common set of dimension tables, forming a fact constellation schema.

- **FactSales:** product-line sales transactions, with measures for revenue, costs, discounts, taxes and profit.
- **FactOrderService:** order-level fulfilment and service information, including delivery performance, returns, cancellations, customer ratings and loyalty programme activity.

4.2.1 Grain

Fact Table	One row represents	Rationale
FactSales	One product line item within a customer order	Sales are recorded per product within an order, so one order can produce several rows. Supports product-level sales and profitability analysis.
FactOrderService	One customer order	Delivery, ratings, returns, cancellations and loyalty activity are recorded per order. A separate table prevents double-counting.

4.2.2 Measures

Fact Table	Additive measures	Non-additive measures
FactSales	Quantity, GrossSales, DiscountAmount, TaxAmount, ShippingCost, NetSales, ProductCost, Profit	UnitPrice, DiscountPercentage
FactOrderService	OrderCount, LoyaltyPointsEarned, LoyaltyPointsRedeemed, ReturnedFlag, CancelledFlag	DeliveryDays, EstimatedDeliveryDays, DeliveryDelayDays, CustomerRating

4.2.3 Keys and Foreign Keys

Each fact table has a surrogate primary key and foreign keys to the dimension tables. OrderID is kept as a degenerate dimension because it identifies orders but does not need a separate dimension table. FactSales has nine foreign keys; FactOrderService has eight because it has no ProductKey, since one order may contain many products.

Fact table	Foreign key	References dimension	Relationship
FactSales	DateKey	DimDate	Many-to-one (fact to dimension)
FactSales	CustomerKey	DimCustomer	Many-to-one (fact to dimension)
FactSales	ProductKey	DimProduct	Many-to-one (fact to dimension)
FactSales	LocationKey	DimLocation	Many-to-one (fact to dimension)
FactSales	SalesChannelKey	DimSalesChannel	Many-to-one (fact to dimension)
FactSales	PaymentKey	DimPayment	Many-to-one (fact to dimension)
FactSales	MarketingKey	DimMarketing	Many-to-one (fact to dimension)
FactSales	ShippingKey	DimShipping	Many-to-one (fact to dimension)
FactSales	OrderStatusKey	DimOrderStatus	Many-to-one (fact to dimension)
FactOrderService	DateKey	DimDate	Many-to-one (fact to dimension)
FactOrderService	CustomerKey	DimCustomer	Many-to-one (fact to dimension)
FactOrderService	LocationKey	DimLocation	Many-to-one (fact to dimension)

4.3 Dimension Tables

Nine dimension tables provide descriptive context for the facts. Each has a surrogate primary key (PK) and supports filtering, grouping, drill-down and roll-up analysis.

Dimension	Attributes	Hierarchy / Notes
DimCustomer	CustomerKey (PK), CustomerID (business key), CustomerName, Age, AgeGroup, Gender, CustomerSegment, City, PostalCode, AcquisitionCost	AgeGroup is derived during ETL.
DimProduct	ProductKey (PK), ProductID (business key), ProductName, ProductCategory, ProductSubcategory, Brand, Supplier, ListPrice, StandardCost, ProductRating	Hierarchy: Category → Subcategory → Product.
DimLocation	LocationKey (PK), Country, Region, StateProvince	Hierarchy: Country → Region → State. Maintained at state level.
DimDate	DateKey (PK), FullDate, DayOfMonth, DayName, WeekOfYear, MonthNo, MonthName, QuarterNo, QuarterName, YearNo, YearMonth, IsWeekend	Hierarchy: Year → Quarter → Month → Date. Includes an Unknown Date member for invalid dates.
DimPayment	PaymentKey (PK), PaymentMethod, PaymentStatus	Supports analysis by payment method and status.
DimSalesChannel	SalesChannelKey (PK), SalesChannel	Compares sales across channels.
DimMarketing	MarketingKey (PK), MarketingChannel, CampaignName, CouponUsed	Hierarchy: Marketing Channel → Campaign. Supports campaign performance analysis.
DimShipping	ShippingKey (PK), ShippingMethod, Warehouse	Supports fulfilment and shipping analysis.
DimOrderStatus	OrderStatusKey (PK), OrderStatus, DeliveryStatus, ReturnStatus, ReturnReason	Junk dimension grouping low-cardinality status attributes.

4.4 Star Schema

The model follows a star schema design with two central fact tables, FactSales and FactOrderService, each connected directly to the shared dimension tables through foreign keys. Because both fact tables share the same dimensions, the design forms a fact constellation, so sales and service performance can be analysed together. DimProduct is used by FactSales only.

Schema Diagram

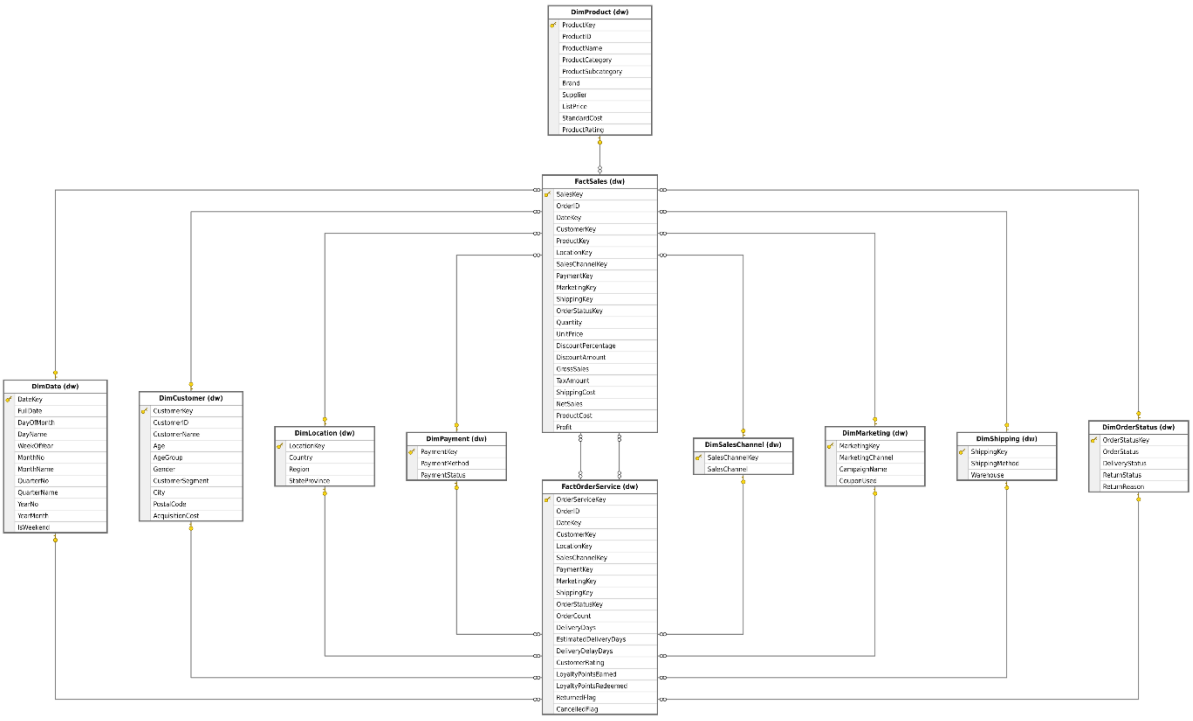


Figure 4.1: Schema Diagram

4.5 Relationships and Hierarchies

All dimension-to-fact relationships are one-to-many, implemented with surrogate keys and foreign key constraints. Each fact row references one dimension row, while one dimension row can relate to many fact rows.

Dimension	Hierarchy
Date	Year → Quarter → Month → Date
Product	Category → Subcategory → Product
Location	Country → Region → State
Marketing	Marketing Channel → Campaign

These hierarchies support drill-down and roll-up in the BI environment, for example from year to quarter, month and date, or from product category to individual products.

4.6 Integration of Data Sources into the Dimensional Model

Data from several source datasets is integrated during the ETL process to populate the fact and dimension tables.

Source file	Used to populate
order_items.csv	FactSales: product-level sales, cost, discount and profit measures
ecommerce_sales_customer_ analytics_150k.csv	FactOrderService and several dimensions: order dates, payment, sales channel, marketing, shipping, delivery, loyalty, ratings, returns and cancellations
customer_master.csv	DimCustomer
product_catalog.json	DimProduct

OrderID links product-level sales records to their order-level information. Customer and product identifiers are validated against their master datasets before loading, and shared dimensions keep both fact tables consistent across all analysis perspectives.

4.7 Design Assumptions

1. Each row in the sales transaction data is one product line item within a customer order, which defines the grain of FactSales.
2. Delivery, ratings, returns, cancellations and loyalty information are stored in FactOrderService at order level to avoid double-counting. FactOrderService has no ProductKey because an order may contain several products.
3. Orders are linked to the customer's location because the source data has no separate shipping-address hierarchy.
4. City and PostalCode are stored in DimCustomer rather than DimLocation because they are too customer-specific to be useful aggregation levels.
5. Missing categorical values (such as campaign, coupon or return reason) are replaced with meaningful labels, for example "No Coupon", during ETL. Missing numeric values remain NULL.
6. Derived attributes (AgeGroup, DeliveryDelayDays, ReturnedFlag and CancelledFlag) are calculated during ETL.
7. Ratios and percentages, such as profit margin and return rate, are calculated in reporting tools or queries rather than stored, because they are non-additive.
8. Surrogate keys are the primary keys of all dimensions; original business identifiers such as CustomerID and ProductID are retained for traceability.
9. Dimensions use a Type 1 Slowly Changing Dimension approach (values are overwritten) because the source systems do not track historical changes.
10. The Date dimension covers all valid order dates in the source data and includes an Unknown Date member (DateKey = -1) for invalid or missing dates.

5. ETL Process Development

5.1 ETL Process Overview

An ETL (Extract, Transform, Load) process was developed in Microsoft SQL Server using T-SQL to move data from the selected CSV sources into the data warehouse. The CSV files are first imported into staging tables in the stg schema. The data is then validated, cleaned, standardized, converted to appropriate data types and integrated, and finally loaded into the dimension and fact tables in the data warehouse schema.

The staging, data warehouse and data mart layers are kept separate using the stg, data warehouse and mart schemas. The SQL script is re-runnable, so the warehouse and data mart objects can be rebuilt consistently.

5.2 Extract

5.2.1 Data Sources

Data Source	Staging Table	Purpose
ecommerce_sales_customer_ analytics_150k.csv	stg.sales	Order, customer, payment, delivery, marketing and service information
customer_master.csv	stg.customer_master	Customer master information
product_catalog.json	stg.product_catalog	Product information
order_items.csv	stg.order_items	Product-level order and sales information
dataset_statistics.csv	stg.dataset_statistics	Reference statistics for validation

The first four files are the main transactional sources. Dataset_statistics.csv is used only as a reference and validation source.

5.2.2 Extraction Method

The CSV files were imported into the EcommerceDW SQL Server database using the SQL Server Management Studio (SSMS) Import Flat File wizard. The data is held in the stg schema, which provides a separate staging area for checking and transforming the data before it is loaded into the warehouse.

5.2.3 Staging Validation

Before transformation, the ETL script compares the columns in each staging table with the expected column names. Any missing required column is returned by the validation query, so the process continues only when the data has been imported with the required structure.

	StagingTable	RowsLoaded	ExpectedRows	Result
1	stg.sales	138116	138116	Pass
2	stg.customer_master	25000	25000	Pass
3	stg.product_catalog	1175	1175	Pass
4	stg.order_items	397569	397569	Pass
5	stg.dataset_statistics	1	1	Pass

Figure 5.1: Staging table validation result

5.3 Transform

The transformation stage prepares the extracted data for loading. It covers data validation, cleaning, standardization, data type conversion, duplicate and missing value handling, derived attribute creation and integration of the source datasets.

5.3.1 Data Cleaning and Standardization

Text attributes are cleaned by removing leading and trailing spaces using LTRIM() and RTRIM(). This includes identifiers such as customer_id, product_id and order_id. Empty text values are replaced with meaningful labels (see Section 5.3.4) so that categorical attributes are consistent in the warehouse.

5.3.2 Data Type Conversion

The source data is converted into appropriate SQL Server data types during the transformation process.

Attribute	Data Type After Transformation
order_date	DATE
customer_age	INT
quantity	INT
delivery_days	INT
estimated_delivery_days	INT
unit_price	DECIMAL(18,2)
gross_sales	DECIMAL(18,2)
net_sales	DECIMAL(18,2)
product_cost	DECIMAL(18,2)
profit	DECIMAL(18,2)
customer_rating	DECIMAL(3,1)

TRY_CONVERT() is used for type conversion so that invalid values can be handled without stopping the complete ETL process.

5.3.3 Duplicate Handling

Duplicates are removed using ROW_NUMBER(), partitioned by the business identifiers customer_id, product_id and order_id for customer, product and order data respectively. Only the first record for each identifier is kept, so duplicate business records are not loaded into the warehouse.

5.3.4 Missing Value Handling

Missing values are handled according to the meaning of each attribute rather than replacing everything with zero.

Missing attribute	Replaced with
Customer name, gender, customer segment (and other unavailable customer or product descriptions)	Unknown
Campaign	No Campaign
Coupon code	No Coupon
Return status	Not Returned
Return reason	Not Applicable

Numeric attributes such as delivery days, estimated delivery days and customer rating are kept as NULL rather than being given an artificial value.

5.3.5 Derived Attributes

Attribute	Derivation
AgeGroup	Derived from customer age when DimCustomer is populated: Missing → Unknown; below 25 → Under 25; 25–34 → 25-34 35–44 → 35-44; 45–54 → 45-54; 55 and above → 55+
CouponUsed	Missing coupon code → No Coupon; existing coupon code → Coupon Used
DeliveryDelayDays	Delivery Days – Estimated Delivery Days. Shows whether an order arrived earlier or later than expected.
ReturnedFlag, CancelledFlag	Boolean indicators of whether an order was returned or cancelled.

5.3.6 Data Integration

The source datasets are integrated using their common business identifiers: customer_id for customer information, product_id for product information, and order_id for order and order-item information. Order-level data from the sales file is combined with product-level data from order_items.

The matching surrogate keys are then found by lookup before the fact tables are populated. The stg.vOrderKeys view performs these lookups for the date, customer, location, sales channel, payment, marketing, shipping and order status dimensions.

5.4 Load

The transformed data is loaded into the dw schema, which contains nine dimension tables (DimDate, DimLocation, DimCustomer, DimProduct, DimSalesChannel, DimPayment, DimMarketing, DimShipping and DimOrderStatus) and two fact tables (FactSales and FactOrderService).

5.4.1 Dimension Table Loading

The dimension tables are loaded before the fact tables so that the surrogate keys exist when fact records are inserted.

Dimension	Populated from
DimDate	Generated for the 2021–2025 order period, plus an unknown-date record for unavailable dates
DimLocation	Unique country, region and state combinations
DimCustomer	Transformed customer data
DimProduct	Transformed product data
DimSalesChannel	Distinct sales channels
DimPayment	Distinct payment method and payment status combinations
DimMarketing	Distinct marketing combinations
DimShipping	Distinct shipping method and warehouse combinations
DimOrderStatus	Distinct order, delivery and return status combinations

5.4.2 Fact Table Loading

Fact Table	Grain (one row represents)	Measures loaded	Notes
FactSales	One product line item within an order	Quantity, Unit Price, Discount Percentage, Discount Amount, Gross Sales, Tax Amount, Shipping Cost, Net Sales, Product Cost, Profit	Created by combining the transformed order_items data with the dimension keys.
FactOrderService	One customer order	Delivery Days, Estimated Delivery Days, Delivery Delay Days, Customer Rating, Loyalty Points Earned, Loyalty Points Redeemed, Returned Flag, Cancelled Flag	A separate order-level table prevents order-level values being counted repeatedly when an order has several product lines.

5.4.3 Loading Sequence

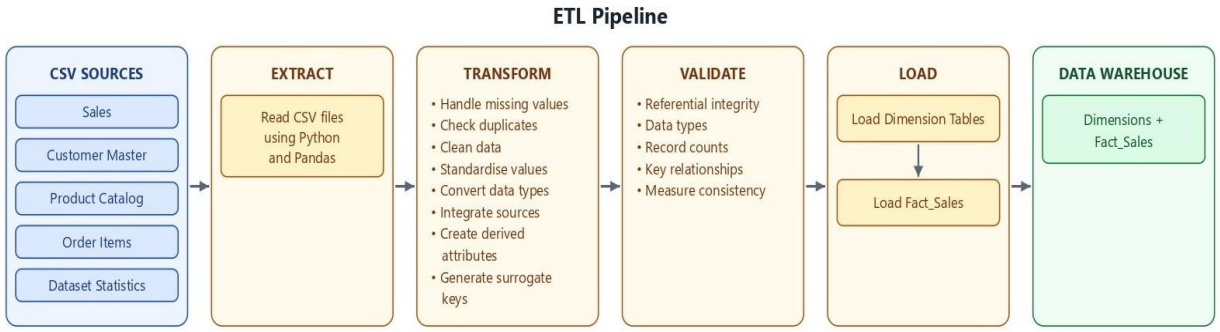
The ETL process follows the sequence below:

Source CSV Files
↓
Staging Tables (stg)
↓
Validation
↓
Transformation Views
↓
Dimension Tables (dw)
↓
Surrogate Key Lookups
↓
Fact Tables (dw)
↓
Validation
↓
Data Mart (mart)

The dimensions are loaded before the fact tables because the fact tables use the dimension surrogate keys as foreign keys.

5.5 ETL Workflow

The complete ETL workflow implemented for the data warehouse is illustrated below.



5.6 ETL Validation Results

Validation checks were run after loading to confirm that the ETL process preserved the required records and measures. The queries compare the source staging tables with the warehouse tables, and compare the sums of key measures between the transformed source data and the fact table. The dataset_statistics source provides an additional reference for the expected values.

Check	Purpose	Source Value	Warehouse Value	Result
Customers	Customer records loaded correctly	25000	25000	Pass
Products	Product records loaded correctly	1175	1175	Pass
Orders	Order records loaded correctly	138116	138116	Pass
Order Items	Product-line records loaded correctly	397569	397569	Pass
Net Sales	Net sales values preserved	192,398,877.29	192,398,877.29	Pass
Profit	Profit values preserved	82,698,038.55	82,698,038.55	Pass
Quantity	Quantities preserved	842366	842366	Pass

Results Messages

	FullDate	YearNo	MonthName	Region	ProductCategory	SalesChannel	MarketingChannel	Quantity	NetSales	Profit
1	2021-11-03	2021	November	North	Books & Media	Marketplace	Organic Search	3	256.58	164.77
2	2022-06-11	2022	June	Central	Office Supplies	Social Media	Email Marketing	1	214.60	132.97
3	2022-06-11	2022	June	Central	Toys & Games	Social Media	Email Marketing	2	368.74	256.18
4	2023-12-11	2023	December	East	Toys & Games	Website	Organic Search	2	189.53	49.12
5	2023-12-11	2023	December	East	Home & Kitchen	Website	Organic Search	1	59.07	15.47
6	2025-07-31	2025	July	Central	Automotive	Mobile App	Affiliate	5	612.65	135.24
7	2025-07-31	2025	July	Central	Office Supplies	Mobile App	Affiliate	2	428.17	232.98
8	2025-07-31	2025	July	Central	Health & Wellness	Mobile App	Affiliate	3	232.37	102.16
9	2025-07-31	2025	July	Central	Fashion	Mobile App	Affiliate	2	471.02	291.31
10	2024-03-05	2024	March	East	Books & Media	Mobile App	Facebook Ads	1	33.44	17.12

	StagingTable	RowsLoaded	ExpectedRows	Result
1	stg.sales	138116	138116	Pass
2	stg.customer_master	25000	25000	Pass
3	stg.product_catalog	1175	1175	Pass
4	stg.order_items	397569	397569	Pass
5	stg.dataset_statistics	1	1	Pass

Figure 5.2: ETL Validation Query Results

6. Data Mart Development

6.1 Data Mart Overview

A department-specific **Sales and Marketing Analytics Data Mart** was developed to provide a business-friendly analytical layer for sales and marketing analysis.

The data mart is created under the mart schema and is based on the dimensional data warehouse. It combines information from the fact and dimension tables into a simplified structure that can be easily accessed by Business Intelligence tools such as Power BI.

The data mart consists of the mart.vSalesAnalytics view, which provides detailed sales and marketing information, and the mart.SalesMonthlySummary table, which provides pre-aggregated monthly analytical data.

6.2 Purpose of the Data Mart

The data mart provides a focused environment for analyzing sales performance, customer segments, products, marketing activities and sales channels, so users do not need to work directly with the underlying warehouse tables. It combines the following:

Component	Content
Dimensions	Date, Customer, Location, Product, Sales Channel, Payment, Marketing, Order Status
Sales measures	Quantity, Gross Sales, Discount Amount, Net Sales, Product Cost, Profit

6.3 Data Mart Structure

mart.vSalesAnalytics is the main analytical view. It combines FactSales with the relevant dimension tables and provides these business-friendly attributes:

Source	Attributes provided
Date	Year, month
Customer	Customer segment, gender, age group, city
Location	Country, region, state
Product	Product category, product subcategory, brand, product name
Sales Channel	Sales channel
Payment	Payment method, payment status
Marketing	Marketing channel, campaign name, coupon usage
Order Status	Order status, return status
FactSales (measures)	Quantity, gross sales, discounts, net sales, product cost, profit

mart.SalesMonthlySummary provides pre-aggregated data at the month × region × product category × customer segment × sales channel level. It contains orders, units sold, gross sales, discounts, net sales and profit.

	COLUMN_NAME	DATA_TYPE
1	SalesKey	bigint
2	OrderID	varchar
3	FullDate	date
4	YearNo	smallint
5	QuarterName	char
6	MonthNo	tinyint
7	MonthName	varchar
8	YearMonth	char
9	DayName	varchar
10	IsWeekend	bit
11	CustomerSegment	varchar
12	Gender	varchar
13	AgeGroup	varchar
14	City	nvarchar
15	Country	nvarchar
16	Region	nvarchar
17	StateProvince	nvarchar
18	ProductCategory	nvarchar
19	ProductSubcategory	nvarchar
20	Brand	nvarchar
21	ProductName	nvarchar
22	SalesChannel	nvarchar
23	PaymentMethod	nvarchar
24	PaymentStatus	nvarchar
25	MarketingChannel	nvarchar
26	CampaignName	nvarchar
27	CouponUsed	varchar
28	OrderStatus	nvarchar
29	ReturnStatus	nvarchar
30	Quantity	int
31	GrossSales	decimal
32	DiscountAmount	decimal
33	TaxAmount	decimal
34	ShippingCost	decimal
35	NetSales	decimal
36	ProductCost	decimal
37	Profit	decimal

Results Messages

	FullDate	YearNo	MonthName	Region	ProductCategory	SalesChannel	MarketingChannel	Quantity	NetSales	Profit
1	2021-11-03	2021	November	North	Books & Media	Marketplace	Organic Search	3	256.58	164.77
2	2022-06-11	2022	June	Central	Office Supplies	Social Media	Email Marketing	1	214.60	132.97
3	2022-06-11	2022	June	Central	Toys & Games	Social Media	Email Marketing	2	368.74	256.18
4	2023-12-11	2023	December	East	Toys & Games	Website	Organic Search	2	189.53	49.12
5	2023-12-11	2023	December	East	Home & Kitchen	Website	Organic Search	1	59.07	15.47
6	2025-07-31	2025	July	Central	Automotive	Mobile App	Affiliate	5	612.65	135.24
7	2025-07-31	2025	July	Central	Office Supplies	Mobile App	Affiliate	2	428.17	232.98
8	2025-07-31	2025	July	Central	Health & Wellness	Mobile App	Affiliate	3	232.37	102.16
9	2025-07-31	2025	July	Central	Fashion	Mobile App	Affiliate	2	471.02	291.31
10	2024-03-05	2024	March	East	Books & Media	Mobile App	Facebook Ads	1	33.44	17.12

Figure 6.1: Data Mart Structure - mart.vSalesAnalytics

6.4 Target Users

The Sales and Marketing Analytics Data Mart is intended for the following users:

- Sales Managers → Sales managers can use the data mart to monitor sales performance, profitability, order volumes, product performance and sales channel performance.
- Marketing Managers → Marketing managers can analyse marketing channels, campaigns, coupon usage, customer segments and their relationship with sales and profit.
- Business Intelligence Analysts → BI analysts can use the data mart as a simplified source for creating reports, dashboards and analytical queries without directly accessing multiple warehouse tables.

6.5 Analytical Benefits

The data mart provides the following analytical benefits:

- **Focused Analysis -** The data mart focuses specifically on sales and marketing information, making it easier for the relevant departments to access the data required for their analysis.

- **Simplified Data Access -** The vSalesAnalytics view combines information from multiple warehouse tables into a single business-friendly analytical layer. This reduces the need for users to write complex joins between fact and dimension tables.

- **Sales Performance Analysis -** Users can analyse gross sales, net sales, profit, quantity and discounts across different products, regions, customer segments and sales channels.

- **Marketing Analysis -** Users can compare marketing channels, campaigns and coupon usage with sales and profitability.

- **Faster Dashboard Analysis -** The SalesMonthlySummary table contains pre-aggregated monthly data, reducing the amount of processing required for common dashboard analysis.

- **Power BI Integration -** The data mart provides a simplified analytical source for Power BI dashboards, allowing business users to interact with sales and marketing information through visual reports.

7. OLAP Analysis and Business Intelligence Dashboard Development

7.1 Dashboard Overview

The Business Intelligence dashboard was developed in Microsoft Power BI Desktop and connected to the EcommerceDW SQL Server database in Import mode. The dw schema tables (FactSales, FactOrderService and the nine dimension tables) were loaded, and Power BI detected the one-to-many relationships between each dimension key and the corresponding fact table foreign key, producing the same constellation structure as the warehouse design. The two fact tables are not related to each other, so sales measures and service measures are always evaluated through the shared dimensions and are not double-counted.

Hierarchies were created for Date (Year, Quarter, Month, Date), Product (Category, Subcategory, Product) and Location (Country, Region, State), and MonthName is sorted by MonthNo. The following DAX measures were defined:

Measure	Definition	Source fact table
Total Net Sales	SUM of NetSales	FactSales
Total Profit	SUM of Profit	FactSales
Profit Margin %	Total Profit ÷ Total Net Sales	FactSales
Total Orders	Distinct count of OrderID	FactSales
Average Order Value	Total Net Sales ÷ Total Orders	FactSales
Return Rate %	Returned orders ÷ all orders (ReturnedFlag)	FactOrderService
Average Rating	Average of CustomerRating	FactOrderService

7.2 Report Page 1: Executive Summary

The Executive Summary page gives senior management a one-screen view of overall performance. KPI cards show Total Net Sales (192.40M), Total Profit (82.70M), Profit Margin (43.0%), Total Orders (138,116), Average Order Value (1,393) and Return Rate (6.9%). Supporting visuals show Net Sales and Profit by month, Net Sales by Product Category, Net Sales by Sales Channel and a Year selector. The headline values reconcile with the ETL validation results (net sales 192,398,877.29, profit 82,698,038.55 and 138,116 orders).

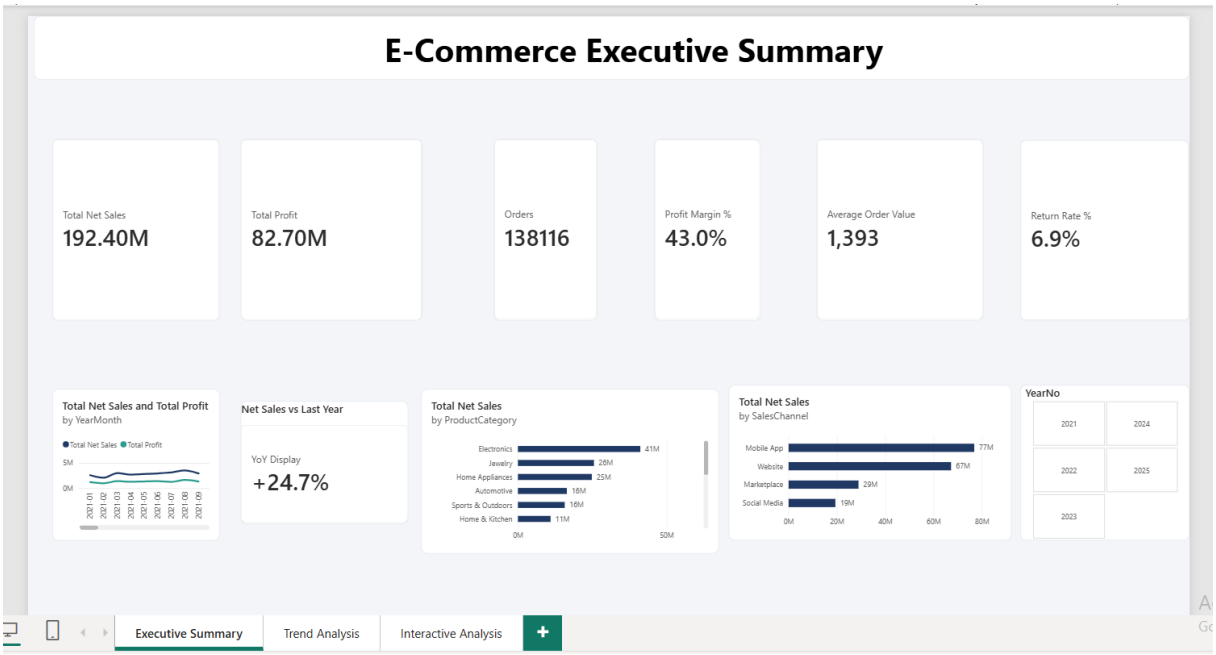


Figure 7.1: Report Page 1 - Executive Summary

7.3 Report Page 2: Trend Analysis

The Trend Analysis page supports time-based analysis and comparison. It contains line charts of Total Net Sales and Total Profit by year-month and of Return Rate % by year-month, a clustered column chart comparing Net Sales by Year and Sales Channel, and a clustered column chart comparing Net Sales by Quarter across the five years.

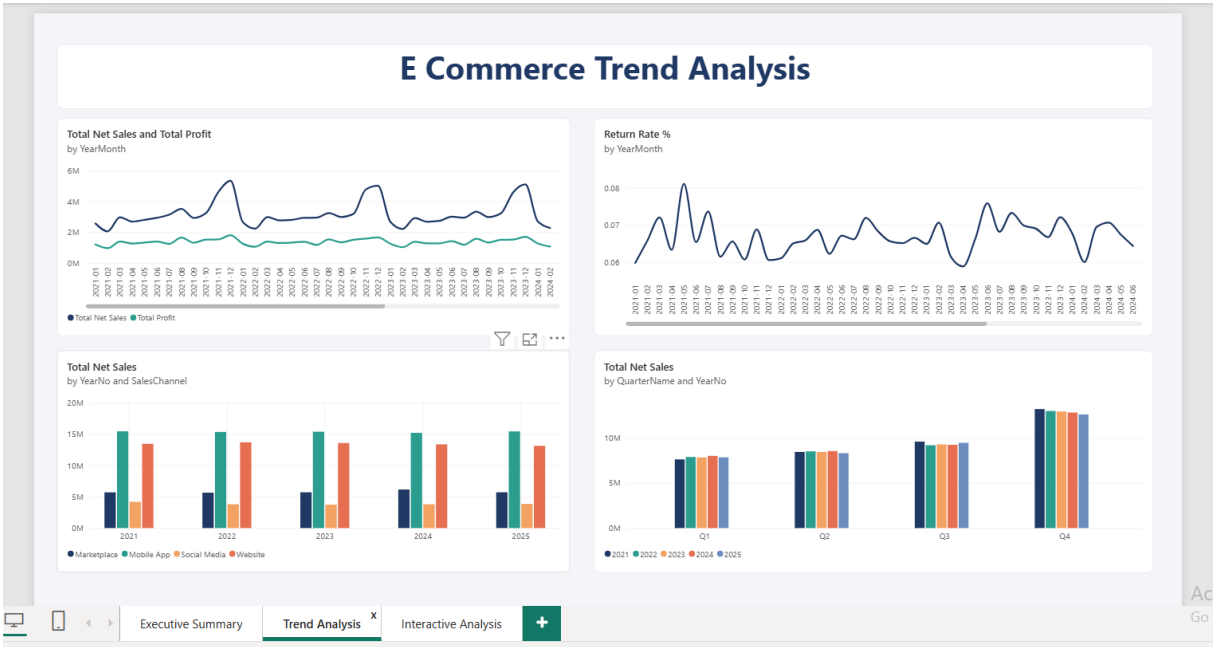


Figure 7.2: Report Page 2 - Trend Analysis

7.4 Report Page 3: Interactive Analysis

The Interactive Analysis page lets users explore the data themselves. Five slicers (Year, Region, Product Category, Sales Channel and Customer Segment) filter every visual on the page at the same time. A matrix with the Product Category, Subcategory and Product hierarchy shows Net Sales, Profit and Profit Margin, and can be expanded and collapsed to drill down and roll up. The Net Sales by Country chart uses the Location hierarchy (Country, Region, State) and the Net Sales by Year chart uses the Date hierarchy, so the drill-down and drill-up controls shown above the visuals move between levels of detail.

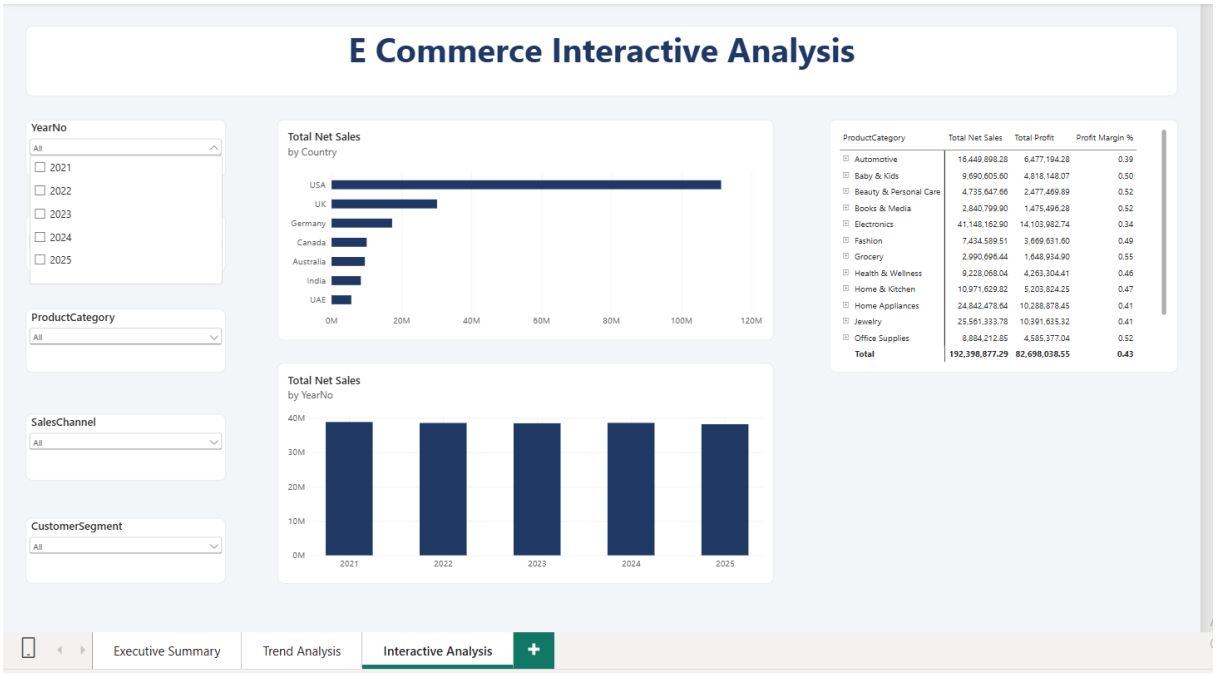


Figure 7.3: Report Page 3 - Interactive Analysis with the Year slicer

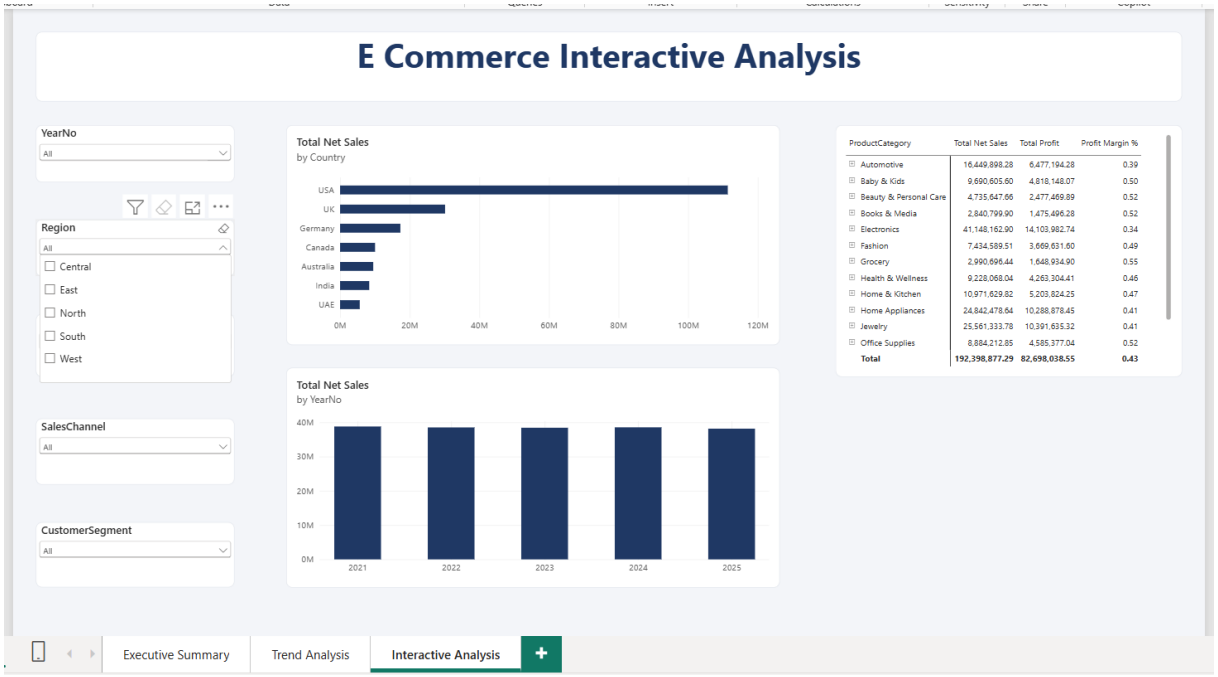


Figure 7.4: Region slicer (Central, East, North, South, West)

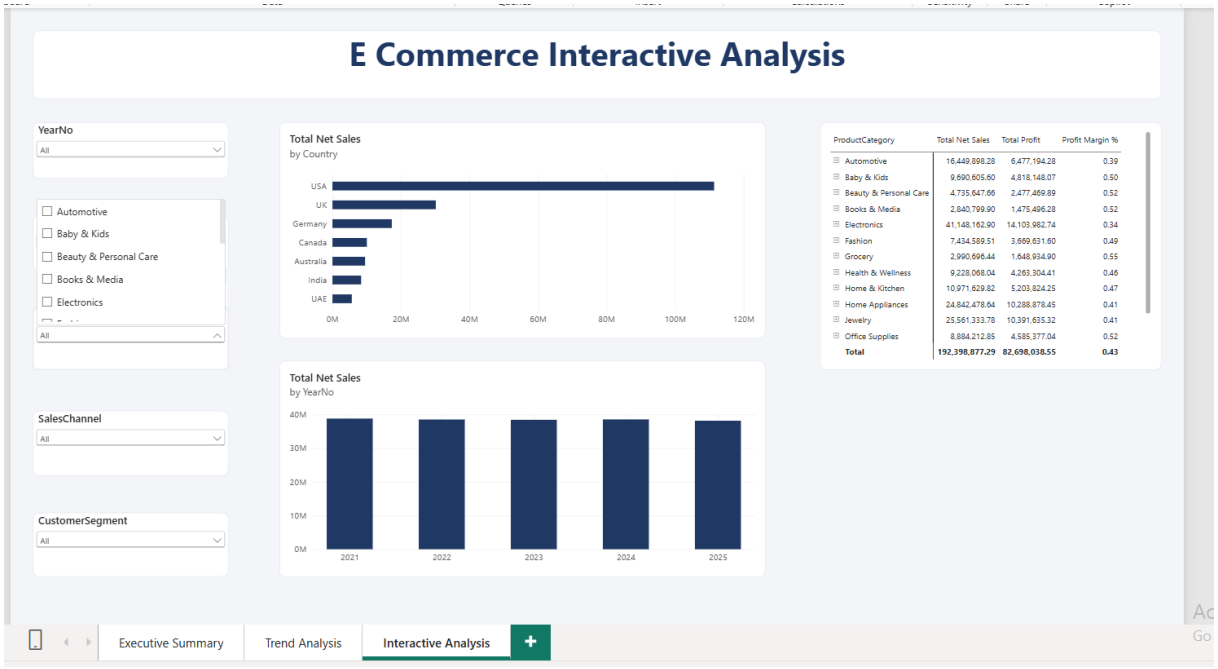


Figure 7.5: Product Category slicer

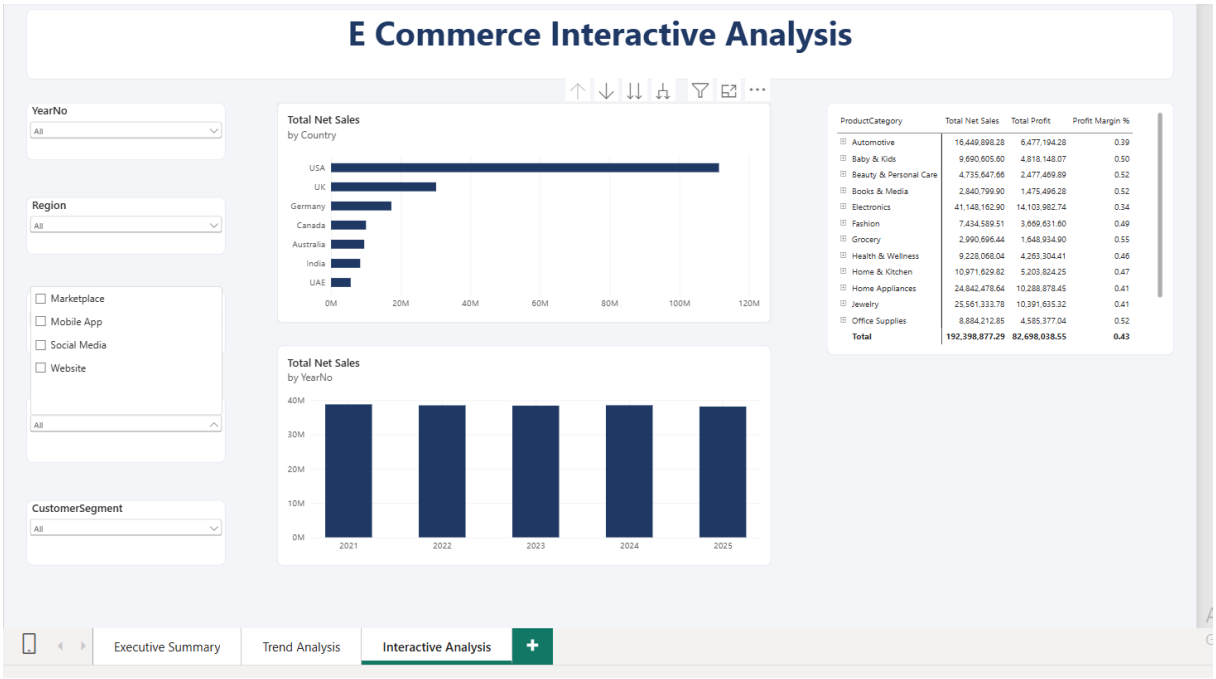


Figure 7.6: Sales Channel slicer, with the drill-down and roll-up controls displayed above the visual

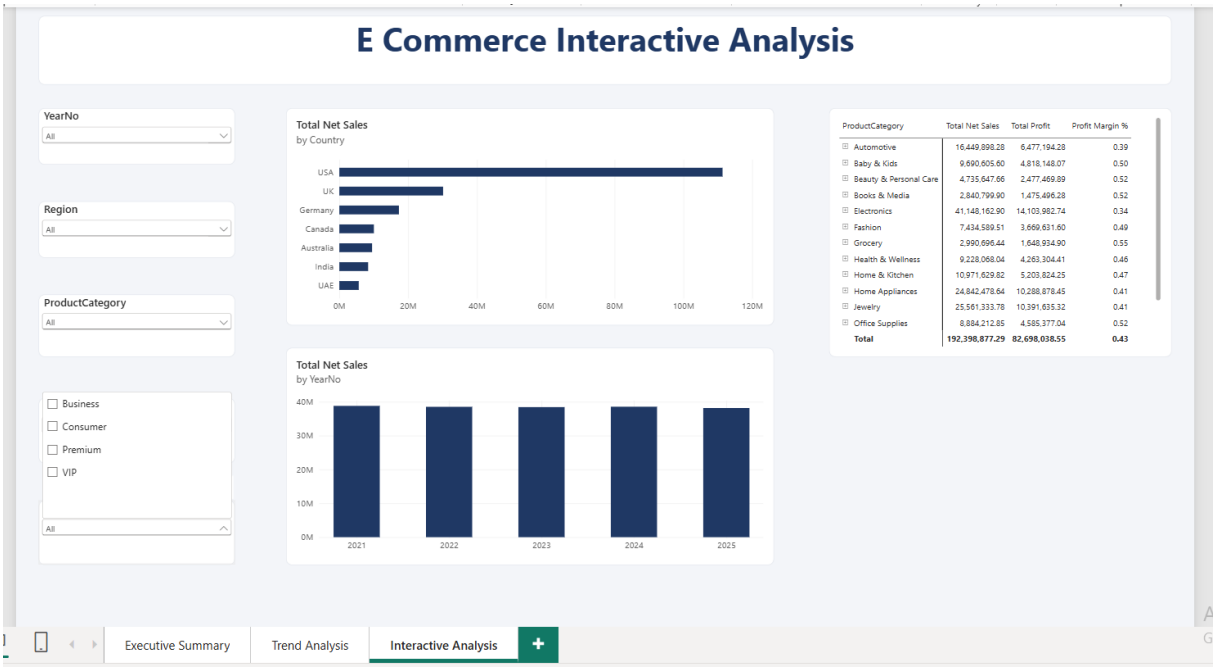


Figure 7.7: Customer Segment slicer (Business, Consumer, Premium, VIP)

8. Business Insights and Recommendations

The insights below are drawn from the dashboard pages in Section 7. Values read from chart axes are approximate; values shown on cards and in the matrix are exact.

8.1 Insight 1: Sales and profit are stable but not growing

Finding: Net sales are almost the same in every year from 2021 to 2025, at roughly 38 to 39 million per year, and the monthly profit line stays between about 1.2 and 1.8 million.

Evidence: Page 3, Net Sales by Year chart; Page 2, Net Sales and Profit by month.

Recommendation: Do not rely on organic growth. Set explicit growth targets and fund them through customer acquisition, retention and loyalty programmes, and track year-on-year growth by segment and channel.

Decision supported: Supports annual budget and target setting, and shows management that the current business has reached a plateau.

8.2 Insight 2: Strong seasonal peak in the fourth quarter

Finding: Q4 net sales are about 13 million in every year, compared with about 7.8 million in Q1, 8.5 million in Q2 and 9.3 million in Q3. Monthly net sales peak at about 4.5 to 5.4 million in November and December against a baseline of about 2.5 to 3 million.

Evidence: Page 2, Net Sales by Quarter and Year chart and the monthly line chart.

Recommendation: Build stock, warehouse and delivery capacity before October each year, schedule the biggest campaigns for Q4, and use targeted promotions in Q1 and Q2 to raise off-peak sales.

Decision supported: Supports inventory purchasing, workforce planning and the marketing calendar.

8.3 Insight 3: Electronics earns the most profit but has the lowest margin

Finding: Electronics has the highest net sales (41.15 million, about 21% of the total) and the highest profit (14.10 million), but its profit margin is only 34% compared with 43% overall. Smaller categories have much higher margins: Grocery 55%, Beauty and Personal Care, Books and Media and Office Supplies 52%.

Evidence: Page 3, Product Category matrix (Net Sales, Profit and Profit Margin %).

Recommendation: Review supplier cost and discounting for Electronics to lift its margin, and promote higher-margin categories through cross-selling and bundles with Electronics purchases.

Decision supported: Supports pricing, supplier negotiation and product assortment decisions.

8.4 Insight 4: Mobile App and Website generate most sales

Finding: Mobile App contributes about 77 million (about 40%) and Website about 67 million (about 35%) of net sales, so around three quarters of sales come from the business's own digital channels. Marketplace contributes about 29 million and Social Media about 19 million, and the shares are almost identical in every year.

Evidence: Page 1, Net Sales by Sales Channel; Page 2, Net Sales by Year and Sales Channel.

Recommendation: Prioritise investment in the mobile app experience and website conversion. Measure the return on Social Media spend before increasing it, and check the margin on Marketplace sales after marketplace fees.

Decision supported: Supports marketing budget allocation and channel strategy.

8.5 Insight 5: Sales are concentrated in the USA

Finding: The USA accounts for about 111 million of the 192.40 million total net sales (roughly 58%). The UK follows with about 30 million and Germany with about 17 million; Canada, Australia, India and the UAE are each below about 10 million, with the UAE lowest at about 5 to 6 million.

Evidence: Page 3, Net Sales by Country chart.

Recommendation: Protect the USA business with reliable delivery and customer service, develop the UK and Germany as the next largest markets, and pilot targeted campaigns in Canada, Australia and India before committing larger investment.

Decision supported: Supports market expansion, logistics planning and risk management for dependence on a single market.

8.6 Insight 6: Returns need continuous monitoring

Finding: The overall return rate is 6.9%, and the monthly return rate moves between about 6% and 8% without a sustained downward trend. The dataset summary file also reports a cancellation rate of 6.08% and an average customer rating of 3.68.

Evidence: Page 1, Return Rate % card; Page 2, Return Rate % by month; dataset_statistics.csv for cancellation rate and rating.

Recommendation: Set a target return rate below 6%, and use FactOrderService to analyse returns by shipping method, warehouse, region and product category to find root causes. Add an operations page to the dashboard for returns, delivery delay and ratings as future work.

Decision supported: Supports quality control, supplier and carrier management, and customer satisfaction improvement.

8.7 Overall Recommendation

Sales are steady but flat, strongly seasonal and concentrated in one country and two digital channels. The business should plan capacity and campaigns around the fourth-quarter peak, improve margin in high-volume categories such as Electronics, grow the secondary markets and channels, and monitor returns and customer ratings so that growth does not come at the cost of service quality.

Acknowledgement of AI Usage

AI tools were used as a supporting tool during this assessment: to help debug SQL import and ETL issues, to explain the Power BI steps for building the model, measures and report pages, and to help structure and check the wording of the dashboard descriptions and insights. The warehouse implementation, the dashboard, the figures and the final conclusions were produced and checked by the group using the project data, and the insight values were verified against the dashboard screenshots.